

Assignment 2

Date of submission: 7th November, 2025

Total marks: 10

Definitions

- **Fair Game:** A game is said to be *fair* if the saddle point is zero, that is, the maximin value equals the minimax value and both are zero.

$$\text{Maximin value} = \text{Minimax value} = 0$$

- **Strictly Determinable Game:** A game is said to be *strictly determinable* if the maximin value equals the minimax value.

$$\text{Maximin value} = \text{Minimax value}$$

Questions

1. For what values of γ , is the game with the following payoff matrix **strictly determinable**? (Ignore the values of γ to determine the saddle point.)

	B_1	B_2	B_3
A_1	γ	6	2
A_2	-1	γ	-7
A_3	-2	4	γ

2. Determine which of the following 2 person zero-sum games are strictly determinable and fair. Give optimal strategy for each player in case of strictly determinable games.

	B_1	B_2
A_1	1	1
A_2	4	3

	B_1	B_2
A_1	-5	2
A_2	-7	-4

Moment generating function

Let X be a random variable. If X is a discrete variable, let $p(x)$ be the probability mass function (pmf) and if X is a continuous random variable, let $f(x)$ be the probability density function (pdf).

If $E(X)$ is the expectation of X , we define the moment generating function (MGF) of X to be

1. $M(t) = E(e^{tX}) = \sum_i e^{tx_i} p(x_i)$, if X is discrete.
2. $M(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$, if X is continuous.

Questions

1. Let X be a random variable and $x_1 = 0$ and $x_2 = 1$ with pmf $p(x_1) = p(x_2) = 1/2$. Find the expectation and moment generating function.
 2. Let X be a continuous random variable with pdf $f(x) = \begin{cases} 2x, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$
Find the expectation and moment generating function.
-

Cumulative Distribution function (CDF)

Let X be a random variable. Then its cumulative distribution function (cdf) is defined by $F_X(x)$, where

$$F_X(x) = P_X((-\infty, x]) = P(\{c \in C : X(c) \leq x\}).$$

We shorten $P(\{c \in C : X(c) \leq x\})$ to $P(X \leq x)$. Also, $F_X(x)$ is often called simply the **distribution function (df)**.

Questions

1. Let X be discrete random variable, if $p(x) = \begin{cases} \frac{x}{15}, & x = 1, 2, 3, 4, 5 \\ 0, & \text{otherwise} \end{cases}$
 - a) Find $p(x = 1 \text{ or } x = 2)$ (**Hint:** Or implies addition)
 - b) $P(\frac{1}{2} < x < \frac{5}{2} \mid x > 1)$ (**Hint:** $P(A \mid B) = P(A \cap B)/P(B)$)
2. If X is a continuous random variable with pdf $f(x) = \begin{cases} C(4x - 2x^2), & 0 < x < 2 \\ 0, & \text{otherwise} \end{cases}$
 - a) find C (**Hint:** Use $\int_{-\infty}^{\infty} f(x) dx = 1$)
 - b) Find $P(x > 1)$ (**Hint:** Use $P(x > 1) = 1 - P(x \leq 1)$)
 - c) Find distributive function.

3. The diameter of an electric cable is assumed to be a continuous random variable with pdf $f(x) = \begin{cases} Kx^3(1-x), & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$
- a) Evaluate K (**Hint:** Use $\int_{-\infty}^{\infty} f(x)dx = 1$)
 - b) Obtain distribution function
 - c) Find $P(X < 2/3)$
4. A continuous random variable X has the pdf $f(x) = 2x, 0 < x < 1$. Find
- i) $P(X < 1/2)$
 - ii) $P(X > 3/4 \mid X > 1/2)$
 - iii) $P(X \leq 3/4 \mid X > 1/2)$